

1 A weighted L^2 estimate for Fourier–GLOF post-processing

Throughout this section, let

$$I = (0, 1), \quad L(x) := -\log x.$$

For $\alpha > -1$ and $\lambda > -1$, define the weighted inner product

$$(u, v)_{\alpha, \lambda} := \int_0^1 u(x) \overline{v(x)} \chi_{\alpha, \lambda}(x) dx, \quad \chi_{\alpha, \lambda}(x) := L(x)^\alpha x^\lambda,$$

and the corresponding norm

$$\|u\|_{\alpha, \lambda} := (u, u)_{\alpha, \lambda}^{1/2}.$$

Let $\alpha, \beta > -1$ and $\lambda > -1$. The generalized log orthogonal functions are defined by

$$\mathcal{S}_n^{(\alpha, \beta, \lambda)}(x) := x^{\frac{\beta-\lambda}{2}} L_n^{(\alpha)}((\beta+1)L(x)), \quad n = 0, 1, 2, \dots,$$

where $L_n^{(\alpha)}$ is the Laguerre polynomial of degree n . These functions satisfy

$$\int_0^1 \mathcal{S}_n^{(\alpha, \beta, \lambda)}(x) \mathcal{S}_m^{(\alpha, \beta, \lambda)}(x) \chi_{\alpha, \lambda}(x) dx = \gamma_n^{(\alpha, \beta)} \delta_{mn},$$

where

$$\gamma_n^{(\alpha, \beta)} = \frac{\Gamma(n + \alpha + 1)}{(\beta + 1)^{\alpha+1} \Gamma(n + 1)}.$$

Let

$$V_M^{\alpha, \beta, \lambda} := \text{span} \left\{ \mathcal{S}_0^{(\alpha, \beta, \lambda)}, \mathcal{S}_1^{(\alpha, \beta, \lambda)}, \dots, \mathcal{S}_M^{(\alpha, \beta, \lambda)} \right\}.$$

The weighted L^2 -orthogonal projection

$$\pi_M^{\alpha, \beta, \lambda} : L_{\chi_{\alpha, \lambda}}^2(0, 1) \longrightarrow V_M^{\alpha, \beta, \lambda}$$

is given by

$$\pi_M^{\alpha, \beta, \lambda} u = \sum_{n=0}^M \widehat{u}_n^{\alpha, \beta, \lambda} \mathcal{S}_n^{(\alpha, \beta, \lambda)},$$

where

$$\widehat{u}_n^{\alpha, \beta, \lambda} = \frac{1}{\gamma_n^{(\alpha, \beta)}} \int_0^1 u(x) \mathcal{S}_n^{(\alpha, \beta, \lambda)}(x) \chi_{\alpha, \lambda}(x) dx.$$

We consider

$$f(x) = a(x) + b(x)x^r L(x)^k, \quad r > 0, \quad k \in \mathbb{N}.$$

Let the Fourier partial sum of f be

$$f_N^F(x) = \sum_{|\ell| \leq N} c_\ell e^{2\pi i \ell x}, \quad c_\ell = \int_0^1 f(x) e^{-2\pi i \ell x} dx.$$

1.1 Assumptions

假设 1 (Basic regularity of the data). *Assume that a, b are analytic in a neighbourhood of $[0, 1]$. In particular,*

$$f \in W^{1,1}(0, 1).$$

Define

$$B_f := \frac{|f(1) - f(0)| + \|f'\|_{L^1(0,1)}}{2\pi}.$$

Under Assumption 1, integration by parts gives, for every $\ell \neq 0$,

$$|c_\ell| \leq \frac{B_f}{|\ell|}.$$

假设 2 (Weighted L^2 GLOF regularization). *Assume that there exists a known sequence $\mathcal{R}_M$ such that*

$$\|f - \pi_M^{\alpha,\beta,\lambda} f\|_{\alpha,\lambda} \leq \mathcal{R}_M.$$

In applications one typically aims to prove

$$\mathcal{R}_M \leq C_{\text{reg}} M^{s_{\text{reg}}} q_{\text{reg}}^M, \quad 0 < q_{\text{reg}} < 1.$$

The role of Assumption 2 is to isolate the regularization part of the proof. It is the part supplied by the GLOF approximation theory. For example, it is verified for finite linear combinations of the model singular functions $x^p(-\log x)^j$ by the single-term GLOF estimate stated below.

1.2 A sufficient condition for the regularization estimate

引理 1 (Single-term GLOF projection estimate). *Let*

$$g_{p,j}(x) := x^p L(x)^j, \quad p \geq 0, \quad j \in \mathbb{N}_0.$$

Assume

$$\beta > \lambda, \quad \lambda > -1 - 2p.$$

Define

$$R_p^{\alpha,\beta,\lambda} := \left| \frac{2p + \lambda - \beta}{2p + 2 + \lambda + \beta} \right|.$$

Then

$$R_p^{\alpha,\beta,\lambda} < 1,$$

and for all sufficiently large M ,

$$\left\| g_{p,j} - \pi_M^{\alpha,\beta,\lambda} g_{p,j} \right\|_{\alpha,\lambda} \leq C_{p,j,\alpha,\beta,\lambda} M^{\frac{\alpha+1}{2}+j} (R_p^{\alpha,\beta,\lambda})^M.$$

命题 1 (Regularization for finite singular expansions). *Suppose that*

$$a(x) = \sum_{j=0}^{J_a} a_j x^j, \quad b(x) = \sum_{j=0}^{J_b} b_j x^j.$$

Assume

$$\beta > \lambda, \quad \lambda > -1.$$

Define the finite set of exponents

$$\mathcal{P} := \{0, 1, \dots, J_a\} \cup \{r, r+1, \dots, r+J_b\},$$

and

$$\rho_{\text{reg}} := \max_{p \in \mathcal{P}} R_p^{\alpha, \beta, \lambda}, \quad R_p^{\alpha, \beta, \lambda} := \left| \frac{2p + \lambda - \beta}{2p + 2 + \lambda + \beta} \right|.$$

Then

$$0 < \rho_{\text{reg}} < 1,$$

and for all sufficiently large M ,

$$\left\| f - \pi_M^{\alpha, \beta, \lambda} f \right\|_{\alpha, \lambda} \leq C_{\text{reg}} M^{\frac{\alpha+1}{2} + k} \rho_{\text{reg}}^M.$$

证明. Since

$$f(x) = \sum_{j=0}^{J_a} a_j x^j + \sum_{j=0}^{J_b} b_j x^{r+j} L(x)^k,$$

we have, by the triangle inequality,

$$\begin{aligned} \left\| f - \pi_M^{\alpha, \beta, \lambda} f \right\|_{\alpha, \lambda} &\leq \sum_{j=0}^{J_a} |a_j| \left\| x^j - \pi_M^{\alpha, \beta, \lambda} x^j \right\|_{\alpha, \lambda} \\ &\quad + \sum_{j=0}^{J_b} |b_j| \left\| x^{r+j} L(x)^k - \pi_M^{\alpha, \beta, \lambda} (x^{r+j} L(x)^k) \right\|_{\alpha, \lambda}. \end{aligned}$$

Applying Lemma 1 to each term gives

$$\left\| f - \pi_M^{\alpha, \beta, \lambda} f \right\|_{\alpha, \lambda} \leq C_{\text{reg}} M^{\frac{\alpha+1}{2} + k} \rho_{\text{reg}}^M.$$

Since the set $\mathcal{P}$ is finite and $\beta > \lambda$, one has

$$R_p^{\alpha, \beta, \lambda} < 1 \quad \text{for each } p \in \mathcal{P},$$

hence $\rho_{\text{reg}} < 1$. □

1.3 The Fourier truncation term

For the truncation estimate, define

$$\eta := \frac{\beta + \lambda}{2}.$$

For each $n \geq 0$, define the GLOF testing kernel

$$W_n^{\alpha, \beta, \lambda}(x) := \mathcal{S}_n^{(\alpha, \beta, \lambda)}(x) \chi_{\alpha, \lambda}(x).$$

Equivalently,

$$W_n^{\alpha, \beta, \lambda}(x) = x^\eta L(x)^\alpha L_n^{(\alpha)}((\beta + 1)L(x)).$$

Let $d \in \mathbb{N}$ satisfy

$$d < 1 + \min\{\alpha, \eta\}.$$

Define

$$A_{n,d}^{\alpha, \beta, \lambda} := \left\| \frac{d^d}{dx^d} W_n^{\alpha, \beta, \lambda} \right\|_{L^1(0,1)},$$

and

$$\mathcal{K}_{M,d}^{\alpha,\beta,\lambda} := \left(\sum_{n=0}^M \frac{(A_{n,d}^{\alpha,\beta,\lambda})^2}{\gamma_n^{(\alpha,\beta)}} \right)^{1/2}.$$

引理 2 (Oscillatory kernel estimate). *Let*

$$K_{\ell n} := \int_0^1 e^{2\pi i \ell x} W_n^{\alpha,\beta,\lambda}(x) dx, \quad \ell \neq 0.$$

If

$$d < 1 + \min\{\alpha, \eta\}, \quad \eta = \frac{\beta + \lambda}{2},$$

then

$$|K_{\ell n}| \leq \frac{A_{n,d}^{\alpha,\beta,\lambda}}{(2\pi|\ell|)^d}.$$

证明. Near $x = 0$,

$$W_n^{\alpha,\beta,\lambda}(x) = O(x^\eta L(x)^{\alpha+n}),$$

and near $x = 1$,

$$W_n^{\alpha,\beta,\lambda}(x) = O((1-x)^\alpha).$$

More generally, for $0 \leq j \leq d-1$,

$$\frac{d^j}{dx^j} W_n^{\alpha,\beta,\lambda}(x) \rightarrow 0 \quad \text{as } x \rightarrow 0^+ \text{ and } x \rightarrow 1^-,$$

because

$$j \leq d-1 < \eta, \quad j \leq d-1 < \alpha.$$

Therefore integration by parts d times produces no boundary terms:

$$K_{\ell n} = \frac{(-1)^d}{(2\pi i \ell)^d} \int_0^1 e^{2\pi i \ell x} \frac{d^d}{dx^d} W_n^{\alpha,\beta,\lambda}(x) dx.$$

Taking absolute values yields

$$|K_{\ell n}| \leq \frac{1}{(2\pi|\ell|)^d} \left\| \frac{d^d}{dx^d} W_n^{\alpha,\beta,\lambda} \right\|_{L^1(0,1)} = \frac{A_{n,d}^{\alpha,\beta,\lambda}}{(2\pi|\ell|)^d}.$$

□

引理 3 (A computable upper bound for $A_{n,d}^{\alpha,\beta,\lambda}$). *Let*

$$b_{n,j}^{(\alpha)} := \frac{(-1)^j}{j!} \binom{n+\alpha}{n-j}, \quad 0 \leq j \leq n.$$

Define coefficients $P_{d,h}(\eta, s)$ recursively by

$$P_{0,0}(\eta, s) = 1,$$

$$P_{d,h}(\eta, s) = 0 \quad \text{if } h < 0 \text{ or } h > d,$$

and

$$P_{d+1,h}(\eta, s) = (\eta - d)P_{d,h}(\eta, s) - (s - h + 1)P_{d,h-1}(\eta, s).$$

Then

$$A_{n,d}^{\alpha,\beta,\lambda} \leq \mathfrak{A}_{n,d}^{\alpha,\beta,\lambda},$$

where

$$\mathfrak{A}_{n,d}^{\alpha,\beta,\lambda} := \sum_{j=0}^n \left| b_{n,j}^{(\alpha)} \right| (\beta+1)^j \sum_{h=0}^d |P_{d,h}(\eta, \alpha+j)| \frac{\Gamma(\alpha+j-h+1)}{(\eta-d+1)^{\alpha+j-h+1}}.$$

Consequently,

$$\mathcal{K}_{M,d}^{\alpha,\beta,\lambda} \leq \left(\sum_{n=0}^M \frac{\left(\mathfrak{A}_{n,d}^{\alpha,\beta,\lambda} \right)^2}{\gamma_n^{(\alpha,\beta)}} \right)^{1/2}.$$

证明. Using the Laguerre expansion

$$L_n^{(\alpha)}(z) = \sum_{j=0}^n \frac{(-1)^j}{j!} \binom{n+\alpha}{n-j} z^j,$$

we obtain

$$W_n^{\alpha,\beta,\lambda}(x) = \sum_{j=0}^n b_{n,j}^{(\alpha)} (\beta+1)^j x^\eta L(x)^{\alpha+j}.$$

A direct induction gives

$$\frac{d^d}{dx^d} [x^\eta L(x)^s] = x^{\eta-d} \sum_{h=0}^d P_{d,h}(\eta, s) L(x)^{s-h}.$$

Therefore

$$A_{n,d}^{\alpha,\beta,\lambda} \leq \sum_{j=0}^n \left| b_{n,j}^{(\alpha)} \right| (\beta+1)^j \sum_{h=0}^d |P_{d,h}(\eta, \alpha+j)| \int_0^1 x^{\eta-d} L(x)^{\alpha+j-h} dx.$$

Since

$$d < \eta + 1, \quad d < \alpha + 1,$$

the integral is finite and equals

$$\int_0^1 x^{\eta-d} L(x)^{\alpha+j-h} dx = \frac{\Gamma(\alpha+j-h+1)}{(\eta-d+1)^{\alpha+j-h+1}}.$$

Substitution gives the claimed bound. $\square$

1.4 Main estimate

定理 1 (Weighted L^2 Fourier-GLOF post-processing error). *Let Assumption 1 hold. Let*

$$\alpha, \beta > -1, \quad \lambda > -1, \quad \eta = \frac{\beta + \lambda}{2}.$$

Let $d \in \mathbb{N}$ satisfy

$$d < 1 + \min\{\alpha, \eta\}.$$

Then

$$\begin{aligned} & \left\| f - \pi_M^{\alpha,\beta,\lambda}(f_N^F) \right\|_{\alpha,\lambda}^2 \\ &= \left\| f - \pi_M^{\alpha,\beta,\lambda} f \right\|_{\alpha,\lambda}^2 + \left\| \pi_M^{\alpha,\beta,\lambda}(f - f_N^F) \right\|_{\alpha,\lambda}^2. \end{aligned}$$

Moreover,

$$\left\| \pi_M^{\alpha,\beta,\lambda}(f - f_N^F) \right\|_{\alpha,\lambda} \leq \frac{2B_f}{d(2\pi)^d} N^{-d} \mathcal{K}_{M,d}^{\alpha,\beta,\lambda}.$$

Consequently, if Assumption 2 holds, then

$$\begin{aligned} & \left\| f - \pi_M^{\alpha,\beta,\lambda}(f_N^F) \right\|_{\alpha,\lambda} \\ & \leq \mathcal{R}_M + \frac{2B_f}{d(2\pi)^d} N^{-d} \mathcal{K}_{M,d}^{\alpha,\beta,\lambda}. \end{aligned}$$

If, in addition, the finite-expansion assumption of Proposition 1 holds, then

$$\begin{aligned} & \left\| f - \pi_M^{\alpha,\beta,\lambda}(f_N^F) \right\|_{\alpha,\lambda} \\ & \leq C_{\text{reg}} M^{\frac{\alpha+1}{2}+k} \rho_{\text{reg}}^M + \frac{2B_f}{d(2\pi)^d} N^{-d} \mathcal{K}_{M,d}^{\alpha,\beta,\lambda}. \end{aligned}$$

证明. Let

$$V_M := V_M^{\alpha,\beta,\lambda}.$$

Since

$$\pi_M^{\alpha,\beta,\lambda} f \in V_M,$$

and $\pi_M^{\alpha,\beta,\lambda}$ is the weighted L^2 -orthogonal projection, we have

$$f - \pi_M^{\alpha,\beta,\lambda} f \perp_{\alpha,\lambda} V_M.$$

On the other hand,

$$\pi_M^{\alpha,\beta,\lambda}(f - f_N^F) \in V_M.$$

Therefore

$$f - \pi_M^{\alpha,\beta,\lambda}(f_N^F) = (f - \pi_M^{\alpha,\beta,\lambda} f) + \pi_M^{\alpha,\beta,\lambda}(f - f_N^F)$$

is an orthogonal decomposition in $L^2_{\chi_{\alpha,\lambda}}(0,1)$. Hence the Pythagorean identity follows:

$$\begin{aligned} & \left\| f - \pi_M^{\alpha,\beta,\lambda}(f_N^F) \right\|_{\alpha,\lambda}^2 \\ & = \left\| f - \pi_M^{\alpha,\beta,\lambda} f \right\|_{\alpha,\lambda}^2 + \left\| \pi_M^{\alpha,\beta,\lambda}(f - f_N^F) \right\|_{\alpha,\lambda}^2. \end{aligned}$$

It remains to estimate the second term. Set

$$e_N(x) := f(x) - f_N^F(x).$$

Then

$$\pi_M^{\alpha,\beta,\lambda} e_N = \sum_{n=0}^M \varepsilon_{n,N} \mathcal{S}_n^{(\alpha,\beta,\lambda)}(x),$$

where

$$\varepsilon_{n,N} = \frac{1}{\gamma_n^{(\alpha,\beta)}} \int_0^1 e_N(x) \mathcal{S}_n^{(\alpha,\beta,\lambda)}(x) \chi_{\alpha,\lambda}(x) dx.$$

Using

$$e_N(x) = \sum_{|\ell|>N} c_\ell e^{2\pi i \ell x}$$

in the L^2 sense, we obtain

$$\varepsilon_{n,N} = \frac{1}{\gamma_n^{(\alpha,\beta)}} \sum_{|\ell|>N} c_\ell K_{\ell n},$$

with

$$K_{\ell n} = \int_0^1 e^{2\pi i \ell x} W_n^{\alpha,\beta,\lambda}(x) dx.$$

By Lemma 2,

$$|K_{\ell n}| \leq \frac{A_{n,d}^{\alpha,\beta,\lambda}}{(2\pi|\ell|)^d}.$$

By Assumption 1,

$$|c_\ell| \leq \frac{B_f}{|\ell|}, \quad \ell \neq 0.$$

Thus

$$\begin{aligned} |\varepsilon_{n,N}| &\leq \frac{1}{\gamma_n^{(\alpha,\beta)}} \sum_{|\ell|>N} |c_\ell| |K_{\ell n}| \\ &\leq \frac{B_f A_{n,d}^{\alpha,\beta,\lambda}}{(2\pi)^d \gamma_n^{(\alpha,\beta)}} \sum_{|\ell|>N} |\ell|^{-d-1}. \end{aligned}$$

Since

$$\sum_{|\ell|>N} |\ell|^{-d-1} = 2 \sum_{\ell=N+1}^{\infty} \ell^{-d-1} \leq \frac{2}{d} N^{-d},$$

we have

$$|\varepsilon_{n,N}| \leq \frac{2B_f}{d(2\pi)^d} N^{-d} \frac{A_{n,d}^{\alpha,\beta,\lambda}}{\gamma_n^{(\alpha,\beta)}}.$$

Using the GLOF orthogonality,

$$\begin{aligned} \left\| \pi_M^{\alpha,\beta,\lambda} e_N \right\|_{\alpha,\lambda}^2 &= \sum_{n=0}^M \gamma_n^{(\alpha,\beta)} |\varepsilon_{n,N}|^2 \\ &\leq \left(\frac{2B_f}{d(2\pi)^d} N^{-d} \right)^2 \sum_{n=0}^M \frac{(A_{n,d}^{\alpha,\beta,\lambda})^2}{\gamma_n^{(\alpha,\beta)}}. \end{aligned}$$

Therefore

$$\left\| \pi_M^{\alpha,\beta,\lambda} (f - f_N^F) \right\|_{\alpha,\lambda} \leq \frac{2B_f}{d(2\pi)^d} N^{-d} \mathcal{K}_{M,d}^{\alpha,\beta,\lambda}.$$

Combining this truncation estimate with Assumption 2 gives the general bound. If Proposition 1 is applicable, substituting its estimate for $\mathcal{R}_M$ gives the final stated inequality. $\square$

1.5 Parameter-dependent form

The previous theorem keeps all parameters explicit. Therefore, for any parameter choice

$$M = M_N, \quad \alpha = \alpha_N, \quad \beta = \beta_N, \quad \lambda = \lambda_N,$$

and for any integer

$$d_N < 1 + \min \left\{ \alpha_N, \frac{\beta_N + \lambda_N}{2} \right\},$$

one obtains

$$\begin{aligned} &\left\| f - \pi_{M_N}^{\alpha_N, \beta_N, \lambda_N} (f_N^F) \right\|_{\alpha_N, \lambda_N} \\ &\leq \mathcal{R}_{M_N}^{\alpha_N, \beta_N, \lambda_N} + \frac{2B_f}{d_N(2\pi)^{d_N}} N^{-d_N} \mathcal{K}_{M_N, d_N}^{\alpha_N, \beta_N, \lambda_N}. \end{aligned}$$

This identity shows explicitly that the truncation term is controlled by

$$N^{-d_N} \mathcal{K}_{M_N, d_N}^{\alpha_N, \beta_N, \lambda_N},$$

where

$$d_N < 1 + \alpha_N \quad \text{and} \quad d_N < 1 + \frac{\beta_N + \lambda_N}{2}.$$

Hence increasing α_N improves the vanishing of the testing kernel at $x = 1$, while increasing $(\beta_N + \lambda_N)/2$ improves the vanishing at $x = 0$. The singularity-matching condition is

$$\frac{\beta_N - \lambda_N}{2} \approx r.$$

A natural parameterization is therefore

$$\lambda_N = \ell N + \lambda_0, \quad \beta_N = \lambda_N + 2r, \quad \alpha_N = aN, \quad M_N = \sigma N,$$

with positive constants a, ℓ, σ and with

$$d_N = \lfloor \theta N \rfloor, \quad 0 < \theta < \min\{a, \ell\}.$$

The remaining nontrivial task is to prove that, under this scaling,

$$\frac{2B_f}{d_N(2\pi)^{d_N}} N^{-d_N} \mathcal{K}_{M_N, d_N}^{\alpha_N, \beta_N, \lambda_N}$$

does not grow faster than the damping factor $N^{-d_N}(2\pi)^{-d_N}$. This is the GLOF analogue of the Fourier–Gegenbauer truncation-error estimate.

注 1 (About standard unweighted $L^2(0, 1)$). *If one takes $\alpha = \lambda = 0$, then*

$$\|\cdot\|_{\alpha, \lambda} = \|\cdot\|_{L^2(0, 1)}.$$

In this case the projection is orthogonal in the standard L^2 norm. However, the endpoint-vanishing condition

$$d < 1 + \min \left\{ \alpha, \frac{\beta + \lambda}{2} \right\}$$

does not allow any positive integer d solely through the above boundary-vanishing argument, because $\alpha = 0$ gives no vanishing at $x = 1$. Thus the present high-order integration-by-parts estimate is naturally a weighted $L^2_{\chi_{\alpha, \lambda}}$ estimate with $\alpha > 0$. In the standard L^2 setting, one still has the stability bound

$$\left\| \pi_M^{0, \beta, 0}(f - f_N^F) \right\|_{L^2(0, 1)} \leq \|f - f_N^F\|_{L^2(0, 1)},$$

but this only gives the usual Fourier-tail rate and does not by itself produce an exponential truncation estimate.